Mathematical overview

In the ideal planar machine, O, X, Q lie on one rigid bar with $OX = XQ = L$; Q, Y, P lie on another with $QY = YP = L$. Short bars XT and TY have length L. On the simple, noncrossed, noncollapsed parallelogram branch $XQYT$, the tracer T and output P satisfy

$$P = O + 2(T - O).$$

This is a dilation of factor 2 about the fixed anchor O. It holds throughout the permitted motion, not just at a few measured poses. With O as origin, take directed half-bars u and v. Then $X = u$, $Q = 2u$, $Y = 2u + v$, $T = u + v$, and $P = 2u + 2v$. Adding directed strips gives a concrete explanation; formal vector notation is optional for children.

For two tracer positions, $P_2 - P_1 = 2(T_2 - T_1)$. Thus all corresponding lengths double, angles and orientation stay the same, and area is multiplied by four. A square of twice the side contains four equal original squares; grid-square and half-square dissections establish the same fact for the packet's polygons. This is not a claim of exact measured copies from an imperfect homemade linkage.

Swapping controlled and drawing points gives the inverse scale $1/2$ and area factor $1/4$. If the tracer picture stays fixed while the anchor moves from O to $O + d$, the new copy is $P' = 2T - (O + d) = P - d$: it moves the same distance in the opposite direction. This comparison requires the same physical input picture and a viable pose for both anchors.

What children discover

Concrete copying and inverse control are an approximate grades 2–3 entry. Upper readiness supports the linkage explanation, area comparison and moving-center investigation. There is no independent K–1 edition: controlled tracing and mechanism recovery are real prerequisites. An adult-led K–1 route follows the launch. Adult reading or scribing is welcome; an adult steering every movement is not independent investigation.

Copies and measurements are experiments. A pattern suggested by several poses is a conjecture. The rigid-bar/parallelogram argument explains every permitted pose. Inaccurate copies may expose loose joints or off-axis pen tips, rather than a counterexample to the ideal theorem.

Limits and status

Prepared, independently reviewed digital prototype; unpiloted. The preparation section names one build, but nobody has built it: the machine, its pencil mounts, clearance, hand access and print fit are unrehearsed. Do not use a crossed or collapsed linkage, and do not describe this packet as hardware-ready until a complete working kit passes the rehearsal on the next page. No statement assumes arbitrary four-bar assemblies have the dilation property.

Preparation and the first visit

Use three pair machines if both older tables participate: two for the four third graders, one for the three older children rotating tracer, observer and referee. The K–1 table follows the K–1 route below. Each pair needs an assembled machine on its base, a joined mat, tracing-paper overlays (one 9×12 inch sheet covers both drawing zones), ruler and straightedge. Adults build and reset the machines.

One build: foam board, brass fasteners and two pencils

Nobody has built this machine yet; it is a plan for the rehearsal below, not a tested kit. Parts for one machine: the four strips on materials page 1, cut from 5 mm foam board; three brass fasteners 20–25 mm long and a few small washers (X, Q, Y); one long pushpin (O); two round coloured pencils about 7 mm thick, blue for T and red for P; a binder clip; and a 20×30 inch foam-board sheet as the base, which meets the 450×400 mm minimum and takes the pin. Rough cost: \$25–40 for three machines, with their bases and a pad of tracing paper.

Glue materials page 1 onto the foam board and cut the strips with a craft knife and a metal ruler; hole centers are at 0, 80 and 160 mm on the long bars and 80 mm apart on the short bars. Make the X, Q, Y holes with a skewer and the O hole with the pushpin itself. Glue an 18×25 mm offcut on top of the P end, flush with it, so that P, like T, holds its pencil in two layers (in one, a tilt moves the point 2–3 mm). Push the pencils themselves through both T holes and both P layers. Stack O–X–Q at the bottom, Q–Y–P and X–T on it, and T–Y on top. Fasten X, Q, Y head down with a washer between the layers, loose enough to turn. The blue pencil through both T holes is the T hinge's axle and the red pencil through both P layers is P's, so each point sits on its pivot axis, as the scale model requires; a pencil taped beside a bar would not. Once the mat is on the base, pin O through it, slide each pencil down to the paper, and clip the binder clip onto a pencil that marks faintly, just above the bar. With pencils at both points, swapping the tips (upper Problem 3) only changes which pencil the child holds.

Foam board needs only a craft knife. If its joints wobble or its holes wear loose in rehearsal, cut the same strips from 3 mm hardboard, use bolts, washers and lock nuts at X, Q, Y, and drill T and P to fit the pencils; this needs a saw and a drill. **Fallback:** a bought drawing pantograph (roughly \$10–25) set to enlarge 2 times and anchored at O. Check that P reaches all four output target corners while T stays inside the input corners, and that T can carry a pencil (shared Problem 2 keeps T's paths; upper Problem 3 swaps the roles). Its bars do not match upper Problem 1's drawing, so that problem becomes an argument about the drawing, with paper strips.

Tape the base to the table. Join the two Letter mat sheets at actual size, matching their registrations and checking the 100 mm segment. Center the joined mat on the base: linkage joints extend outside the paper. Preserve full-zone support and hand clearance. Put fresh overlays on the same registration, retaining the fixed anchor and earlier copies.

Required physical rehearsal (not yet performed)

Assemble one machine. Confirm a simple open rhombus and the midpoint topology. Put T at nine positions spanning the marked zone and check O, T, P alignment and $OP = 2OT$. Trace straight, cornered and closed paths; exchange drawing/control roles; test the alternative anchor if that route will be used. Check that neither point lifts, drags or crosses branches, that both pencils mark the tracing paper without pressing, that each pencil point moves no more than about 1 mm when the pencil is pushed sideways, that the frame stops before it folds or collapses (Shared Problem 3 below), that all bars stay supported, and that two children's hands fit. Reject loose or easily crossing assemblies. Mark the intended branch; an adult resets any stuck/collapsed assembly. A screen animation does not satisfy this rehearsal.

Budget about 30 minutes to build the first machine and 30–60 minutes to rehearse it, then about 20 minutes for each further machine and 10–15 minutes to print and join each mat. These are untested estimates. If the mechanism fails, postpone its student tasks and keep the mathematics for a later visit.

Launch together

After the customary five-minute run, let children handle the mechanism gently with tips raised. Anchor O and trace one small curved stroke with T, pausing midway to show the emerging P stroke. Keep the same anchor and input through the completed demonstration. Say: "You choose the tracer's path. The other pencil makes its copy. Stop if the bars get stuck." Give each child a turn before distributing tasks. Leave scale and area discoveries for their attempts.

K–1 route

There is no K–1 packet, and the K–1 table does not wait for the machine. After a machine has passed the rehearsal, an adult can offer a short station, one child at a time: the adult steadies the base and keeps O pinned, the child traces insert A and then C with T, and the adult asks, "What did the red pencil draw?" A child who wants more steers T so that P touches every dot on target D, in any order. Plain-words key: the red drawing is the same shape, pointing the same way, only bigger, with every part twice as long; to move P toward a dot, T moves the same way but half as far. That K–1 children can steer T while watching P is a prediction. Before the station, or instead of it, use a hardware-free K–1 packet such as the Week 1 encore's K–1 Problem 1, which fills the big pattern blocks with small ones of the same colour.

Shared packet: copying and control

The shared packet has four problems. A first visit can explore Problems 1–2 for most of the table time, then compare copies. A suggested first hour, untested: the five-minute run, a 10-minute launch with a turn for each child, about 30 minutes on Problems 1–2, and 10 minutes for Problem 3 or 4 and comparing copies. Problems 3–4 can follow when children want a rule or another control challenge. Give only the inserts for the current task and keep one useful overlay for the next visit. A separate upper packet has its own Problems 1–4.

Shared Problem 1: paths A, B, C

Every stroke of the output is the same shape and orientation at twice the scale, positioned by dilation about O. In A, 35 mm then 20 mm becomes 70 mm then 40 mm. In B, the triangle's corresponding sides double; it does not rotate or turn over. In C, the staircase segment lengths 10,15,20,25,10 mm become 20,30,40,50,20 mm. Corners and the order of the path are retained. Several repeated trials are useful, especially if corners introduce pen drag.

Hold back measuring instructions until copies exist. If useful, offer one strip to compare corresponding segments or ask which tip drew the larger picture. An apparent rotation, reflection or varying scale prompts an anchor/branch/mount check, not an immediate correction of the child's observation. The ideal claim and the real machine's precision are different questions.

Shared Problem 2: visit D and trace E

Controlling T while watching P is an inverse task. With $O = (40, 140)$ in mat millimeters, the inverse is $T = (P + O)/2$. The five D target dots correspond to tracer positions $(115, 125)$, $(130, 160)$, $(155, 150)$, $(150, 120)$, $(125, 140)$. Children may visit them in any order; the task is control, not a prescribed route.

E's house vertices correspond to $(120, 125)$, $(150, 125)$, $(150, 145)$, $(135, 157.5)$, $(120, 145)$, then return. Children need not use coordinates or fractions: the two tips trace both paths physically. Give a fresh overlay when the first path becomes confusing. A held hint is to place T tentatively and observe which direction P moves, preserving the fixed anchor.

Shared Problem 3: a rule for poses

No permitted ideal pose breaks O, T, P collinearity or $OT = TP$ (equivalently $OP = 2OT$). Children can make experiments with a straightedge, string or reusable strip if they ask how to compare. Finite testing suggests a conjecture; the upper packet's bar argument establishes it for every permitted pose. Do not count a stuck, crossed or off-axis-tip assembly as a counterexample to these assumptions. On a real frame two poses lie outside them. At full stretch ($OT = 160$ mm) T reaches Q and the short bars can fold onto the long ones: then $OT = TP = 160$ mm still, but O, T, P need not lie on one line. At full collapse Y meets X, and T and P meet O. In the foam-board build the strips should stop the frame before either pose (a rehearsal check): X–T and Q–Y share a layer and meet when OT rises above about 159 mm, and the offcut glued on the P end meets T–Y, the top strip, when OT falls below about 22 mm. The T zone keeps OT between 70 and about 123 mm, well clear of both.

Shared Problem 4: where T can go

F is a 40×30 mm output rectangle with corners $(220, 125)$ and $(260, 155)$. Since $T = (P + O)/2$, P is inside F exactly when T is inside the half-size rectangle with vertices $(130, 132.5)$, $(150, 132.5)$, $(150, 147.5)$, $(130, 147.5)$: 20×15 mm, the same shape and orientation, half as far from O. Every T in it sends P into F, and every P in F comes from a T in it, so this rectangle is the whole region, and its edge is the path T draws while P follows F's outline. Children may first mark separate points, pushing P against each side of F and marking where T must stop, then join them. Plain-words answer: a smaller box, half as wide and half as tall, closer to O. A held question after the edge is marked: how do the two boxes compare?

Upper packet: explanation and return visits

Suggested pacing, untested: each upper problem is a 15–25 minute investigation, and one or two of them make a return visit.

Upper Problem 1: why doubling persists

Name the equal directed half-bars $u = \overrightarrow{OX} = \overrightarrow{XQ}$ and $v = \overrightarrow{QY} = \overrightarrow{YP}$. In the intended parallelogram, $\overrightarrow{XT} = v$ and $\overrightarrow{TY} = u$. Thus $\overrightarrow{OT} = u + v$ and $\overrightarrow{OP} = 2u + 2v = 2(u + v)$. Loose directed paper strips or arrows on the diagram can show this without symbolic algebra. A page of measurements cannot replace that rigid-frame explanation. Equal-length marks alone do not specify the operating branch: keep the same simple rhombus as the real assembly.

Upper Problem 2: area of G, H, I

The 10 mm square is one area unit. G has area 4 units and its copy 16; H also has area 4 and its copy 16; I is the right-isosceles triangle with 30 mm legs, area $4\frac{1}{2}$, and its copy 18. These are square units, not boundary lengths or raw piece counts. Two half-square triangles make one unit. The triangle's diagonal aligns with the supplied half-square pieces, so exact covering is possible in the ideal drawing. Cover one shape/copy at a time and reuse the kit; it is not necessary to leave all six covered simultaneously.

A held hint is to place four unit squares into a doubled unit square. A grid-square dissection of G and H then gives four copies of every original unit. For I, pair two congruent triangles into a square, use the fourfold square area, then take half. This explains all three and extends to polygons decomposed into these units. Physical tracing thickness and play in joints may prevent exact edge matching; compare the intended geometry with the measured result.

Upper Problem 3: reverse the machine

Children now move P and T makes the reduced drawing; with the foam-board build an adult only says which pencil to hold. J's 60×40 mm rectangle becomes 30×20 mm. Every corresponding arrow length in K halves, retaining angles and orientation. The full relation is now $T = O + \frac{1}{2}(P - O)$. To make T fill L, whose bounds are $(125, 130)$ to $(155, 150)$, P must trace the 60×40 mm rectangle from $(210, 120)$ to $(270, 160)$. If the mount does not permit a reliable exchange of roles, postpone this visit rather than silently moving an off-axis pencil.

Upper Problem 4: change the anchor

The printed second anchor $O' = (60, 140)$ is 20 mm to the right of O. Keeping M fixed moves its copy 20 mm left: $P' = 2T - O' = P - (20, 0)$. To make the second copy land on the first, move every point of M 10 mm right, because $2(T + (10, 0)) - O' = 2T - O$. Its new tracer vertices lie within the intended zone. The shape and area do not change. The demonstration must retain M's physical location while the anchor changes for the first comparison; shifting M at the same time would answer a different question.

The general compensation is half the anchor displacement in the same direction. Ask for a new displacement only after the first two copies are visible; no simultaneous coordinate table is needed. Retain both overlays and the anchor labels as a recoverable record.

Print selection, provenance and use record

The shared and upper packets are independent consecutively numbered routes. Use the matching lettered materials, not rescaled thumbnails, on the joined mat. The materials include adult templates and setup, the two mat halves, seam bridge, inserts A–M, and area pieces. The source README gives the exact current material page map. Print the cutting template only for adult building; children use assembled, rehearsed machines.

Common failure checks

If an insert seems to demand a stuck pose, stop. Confirm the named insert's registration, correct control point, anchor position and intended branch before trying again. Do not force a joint. Tip offsets or loose pivots can enlarge errors even when bar-hole distances are right. Joined paper may still be too small to support the outer joints; the base is required. Adults should not operate both tips to produce an attractive result on a child's behalf.

Sources and adaptation

Glen Whitney, *Math Monday: Linkages – The World "B.X."*, National Museum of Mathematics, Pantograph ingredient/directions/use paragraphs (first appeared October 8, 2012): two midpoint long bars, two half-length bars, fixed endpoint, double/half tracing. See the MoMath article (full locator in SOURCES.md). The millimeter kit, particular inserts, operating regions, anchor comparison and proof presentation are locally authored adaptations. No source photograph establishes this kit's physical readiness.

Givental, Nemirovskaya and Zakharevich, *Math Circle by the Bay*, Preface viii–x, supports deep revisitable themes, manipulatives and varying pace. The particular routes, timing and hardware estimates here are editorial judgments, not results reported by that book. The recovered Week85 plan and production record preserve provenance and review history.

After a visit

Record which packet/problem/inserts children actually used, whether they controlled a point independently, what comparisons they noticed, where the mechanism stuck, and which question they wanted next. Separate those observations from hypotheses about motor control or explanations. Save one identified overlay and a picture of the working assembly for a return visit. Prepared pages alone are not taught material.

All outputs remain unpiloted; exact fit, operation and hand clearance are physically untested. Digital review and rebuilding check the published artifacts, not the real equipment.